

6.4 Selecting Events

1. Cut Flow Histogram

We apply a series of cuts to our data and create a cut-flow histogram to show how much data is being removed/kept at each step.

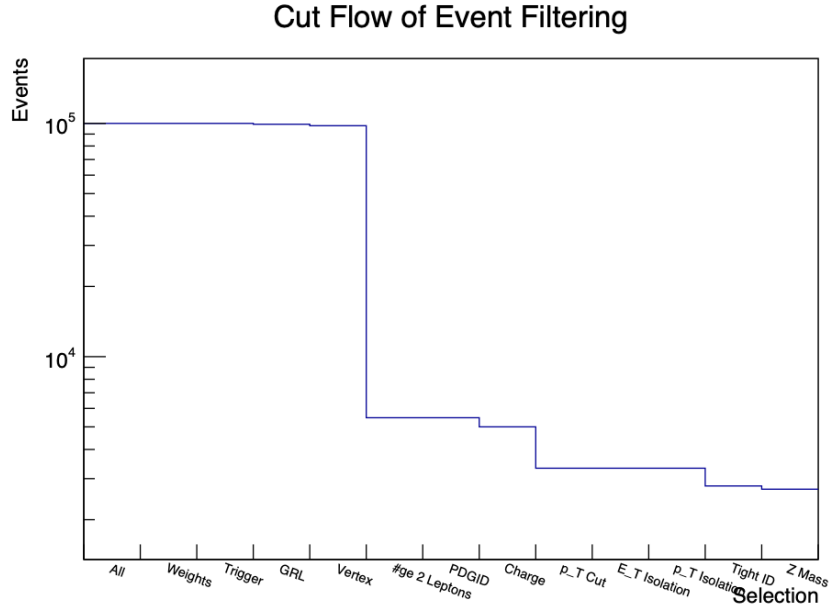


Figure 1: Cut Flow Histogram

List of cuts: - Weights: MC weighting - Trigger: Check `trigE` or `trigM` is true (electron or muon trigger from dataset) - GRL: Good run list, checks if event is marked as a good run (Data Quality Group assessment) - Vertex: Check `hasGoodVertex` variable, meaning primary vertex/position of collision is well determined. - ≥ 2 Leptons: Check that at least two leptons were detected (otherwise it is not a Z decay event) - PDGID (Particle Data Group Identifiers): check `lep_type` variable and make sure the measured particles are leptons (e, μ, τ). - Charge: Z Boson is neutral so the decay products must have opposite charges. - p_T Cut: leptons must have at least 25 GeV of transverse momentum (leptons are expected to split their momenta to half of the Z mass), cutoff below this. - Isolation Cut (p_T): see below - Isolation Cut (E): see below - Tight ID: We want to have data with harsher identification criteria - Z Mass Cut: see below

The most effective cut is the cut requiring at least two leptons be produced and measured in the event.

Next most effective is the p_T cut.

Isolation cut:

We want events *without* other particles around our collision products. This means we should cut out events which have other energy/momentum in the same region of the detector where the lepton passed through.

This cuts out events where the lepton might have interacted with other particles and *lost energy to surroundings*, because that would not be accounted for in our calculation.

We create a histogram of p_{Tcone}/p_T to see its distribution and to select a reasonable cutoff.

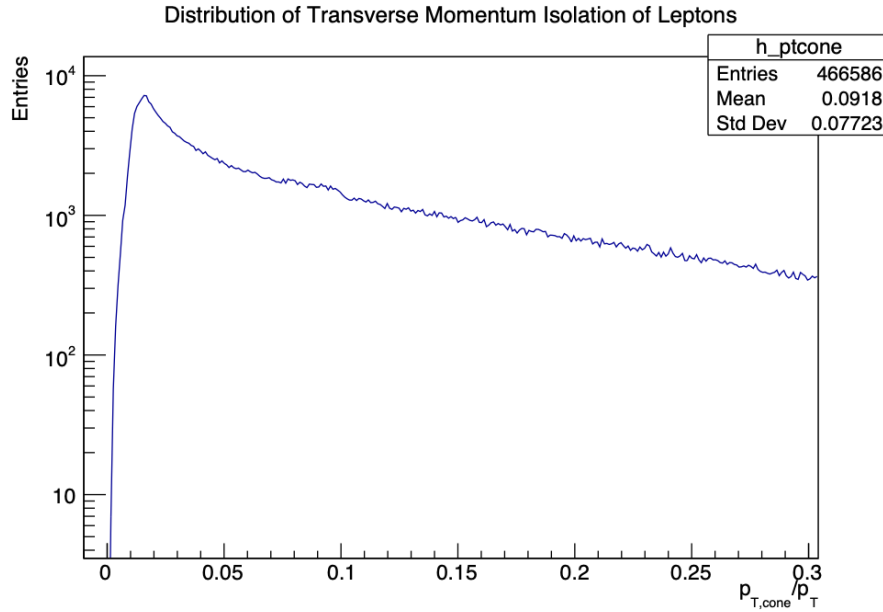


Figure 2: pt Cone Ratio

We restrict $Etcone20 / E$ and $ptcone30 / pt$ to ≤ 0.13 as our cutoff limits.

M_Z cut:

We restrict the invariant mass to a reasonable range:

$$60 \text{ GeV}/c^2 \leq M_Z \leq 120 \text{ GeV}/c^2$$

3. Applying to Datasets

We run the cutting process on all the events on the following files, creating invariant mass histograms for all of them.

- DataEgamma.root
- DataMuons.root
- mc_147770.Zee.root
- mc_147771.Zmumu.root
- mc_147772.Ztautu.root

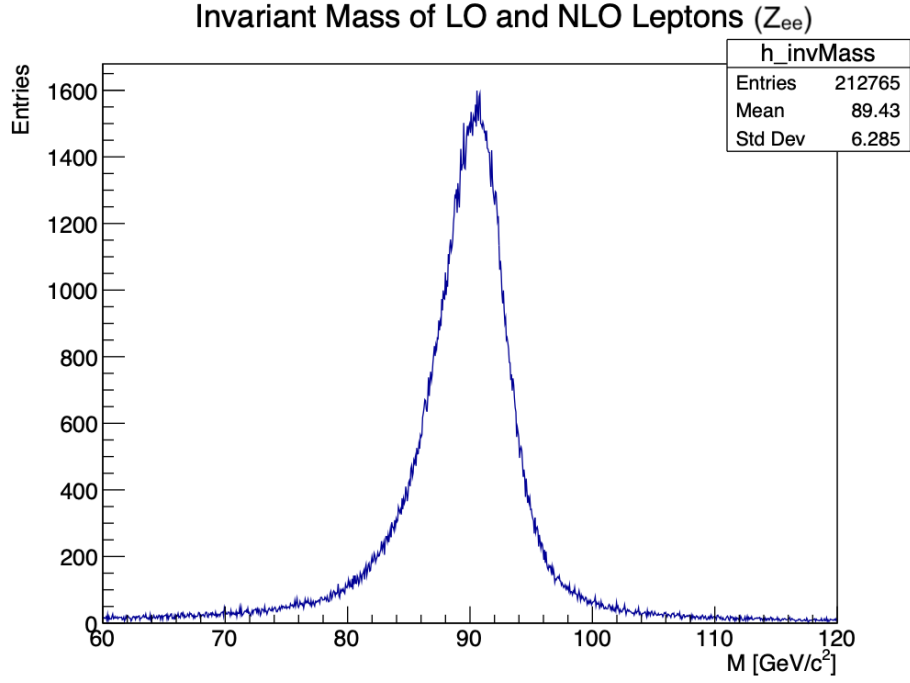


Figure 3: DataEgamma Graph of Invariant Mass

The last graph clearly does not show have a peak at the expected ~ 90 GeV. This is because the taus produced in a Z decay can themselves decay very quickly in many ways:

$$\tau^- \xrightarrow{W^-} \ell^- + \bar{\nu}_\ell + \nu_\tau$$

$$\tau^- \xrightarrow{W^-} \text{hadrons} + \nu_\tau$$

And for anti-tausons:

$$\tau^+ \xrightarrow{W^+} \ell^+ + \nu_\ell + \bar{\nu}_\tau$$

$$\tau^+ \xrightarrow{W^+} \text{hadrons} + \bar{\nu}_\tau$$

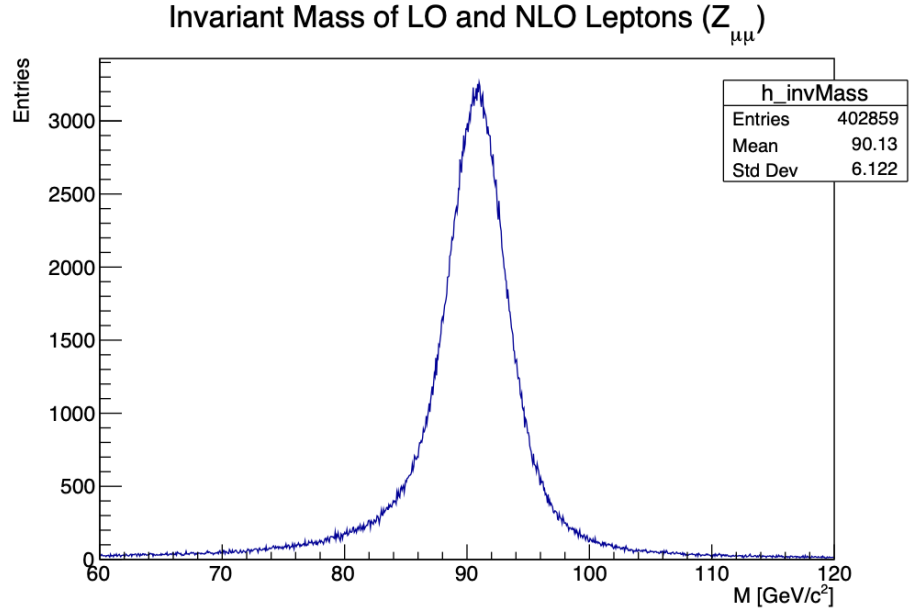


Figure 4: DataMuons Graph of Invariant Mass

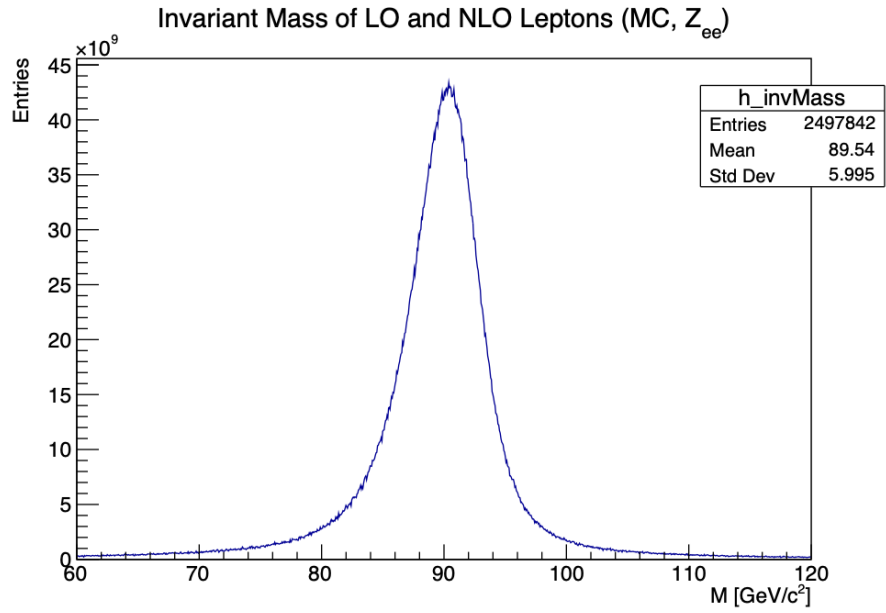


Figure 5: MC Electrons Graph of Invariant Mass

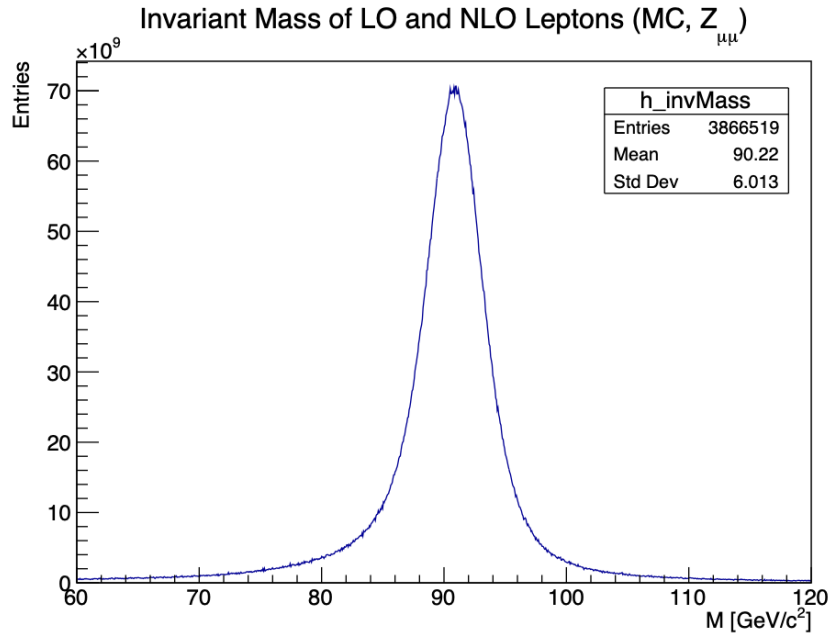


Figure 6: MC Muons Graph of Invariant Mass

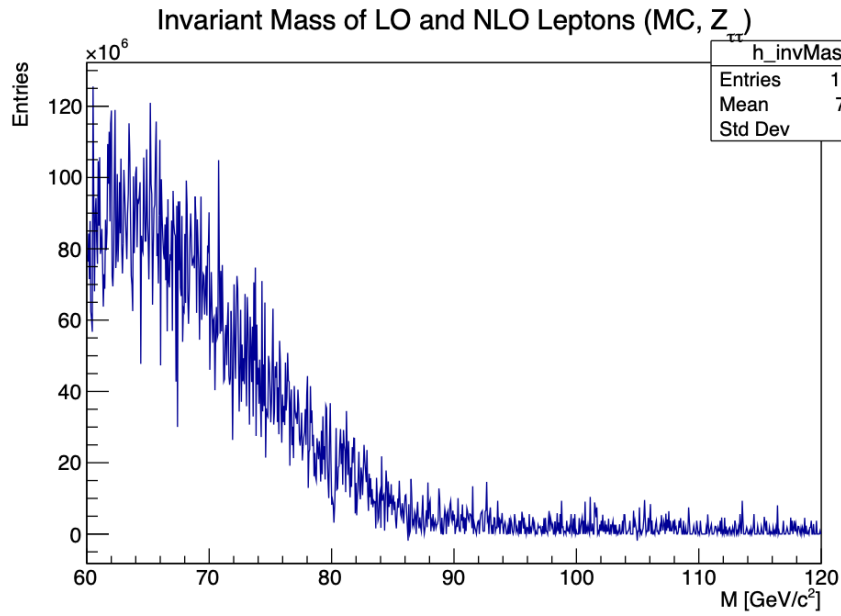


Figure 7: MC Tauons Graph of Invariant Mass

(about 17% per lepton branch, 65% decay into hadrons)

Firstly, we are only looking at lepton decays. Filtering out the most likely decay means we are losing a lot of tauon events.

For the decays where Z bosons become $\tau^\pm$, these can decay again into the above leptons. Because neutrinos are not detected, we do not measure the total energy/mass/momentum. Without the neutrinos, the reconstructed Z candidate has too low mass.